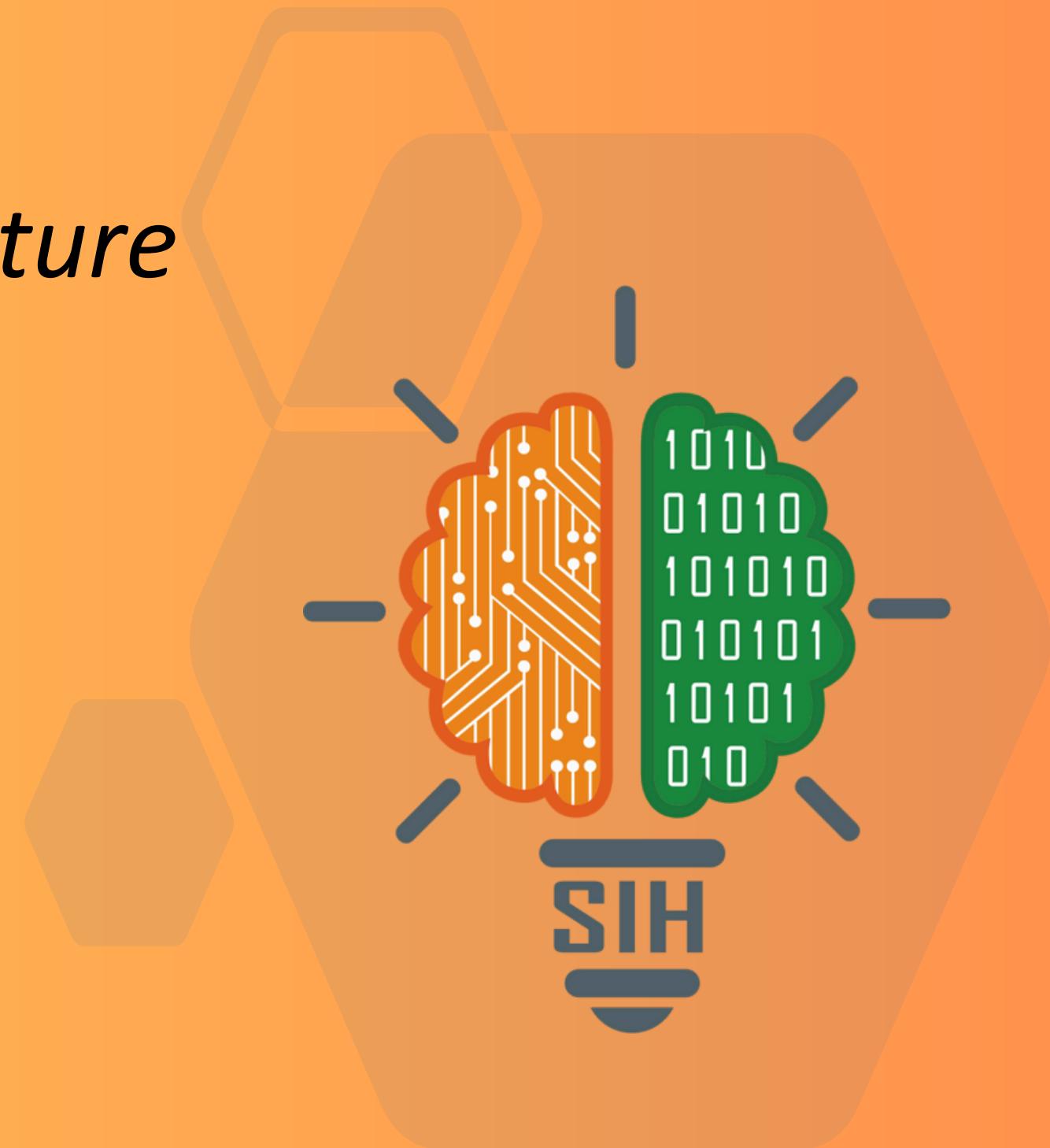


TITLE PAGE

- **Problem Statement ID** –*SIH25062*
- **Problem Statement Title**-*Smart Agriculture in hilly regions*
- **Theme**-*Agriculture, FoodTech & Rural Development*
- **PS Category**-*Hardware*
- **Team ID**-
- **Team Name**-*SLASHA*



Problem statements

- Traditional irrigation fails to meet crop needs during dry summers.
- Inadequate water management leads to frequent water scarcity and loss.
- Lack of smart technology limits irrigation efficiency and resilient farming.

Innovation and Uniqueness

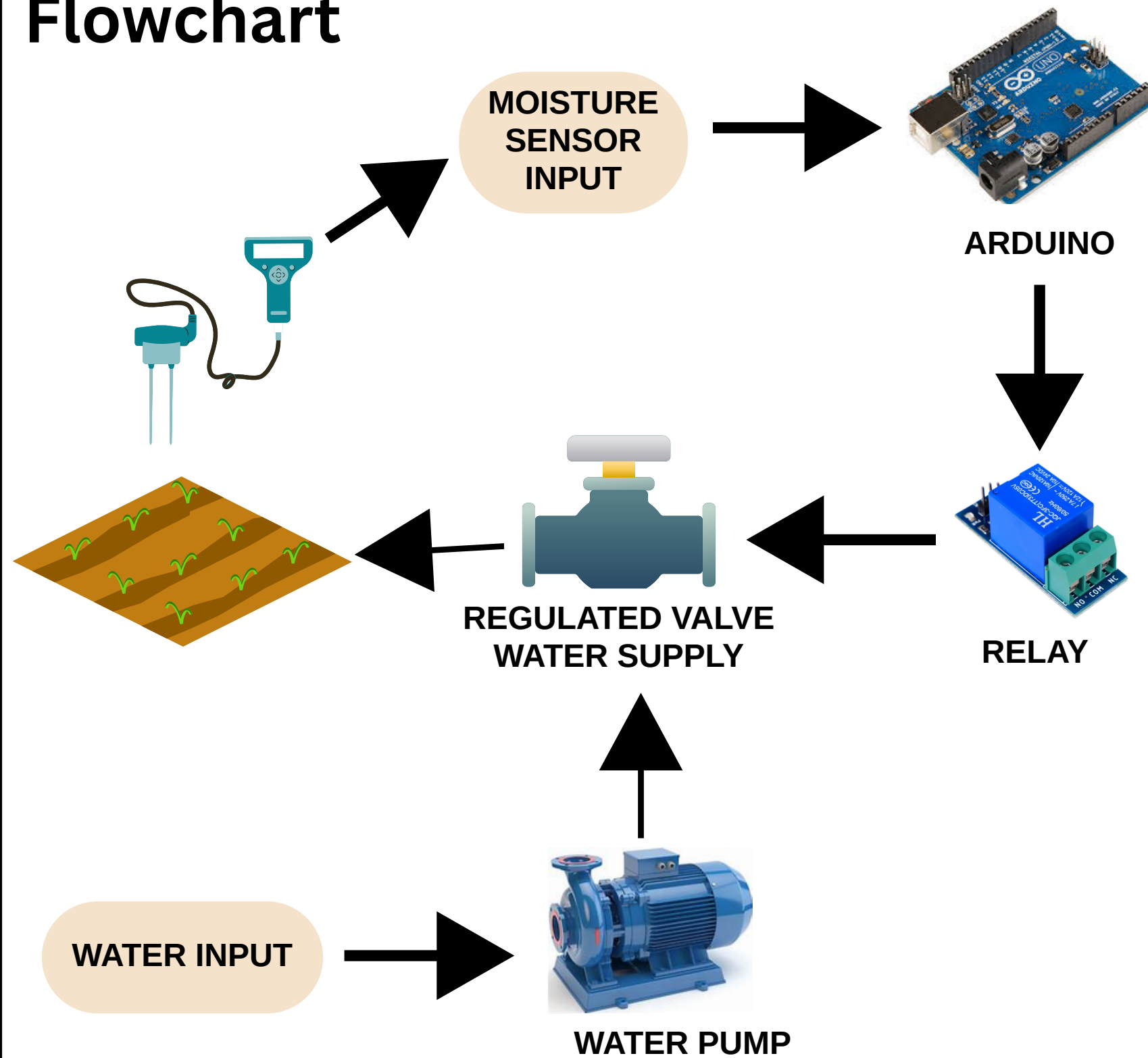
- Installation of power sources like **solar** and **windmills** to power the pumping.
- Using **frost protector** for crops to protect them from **snow fall**.
- Implementing **step farming** to reduce nutrient runoff.

Idea/Solution

Implementation of **A Sensor-Based Smart Irrigation System** with Crop Intelligence and **Rainwater Harvesting**.

- Continuous monitoring of field conditions using **soil moisture** and **temperature sensors**.
- **Automated watering** system using **microcontrollers** by reading field crop database.
- A **mobile/web dashboard** with **SMS alerts** to help farmers monitor and control irrigation remotely.
- **Rainwater harvesting** system integrated with **level sensors** for real-time water level tracking.

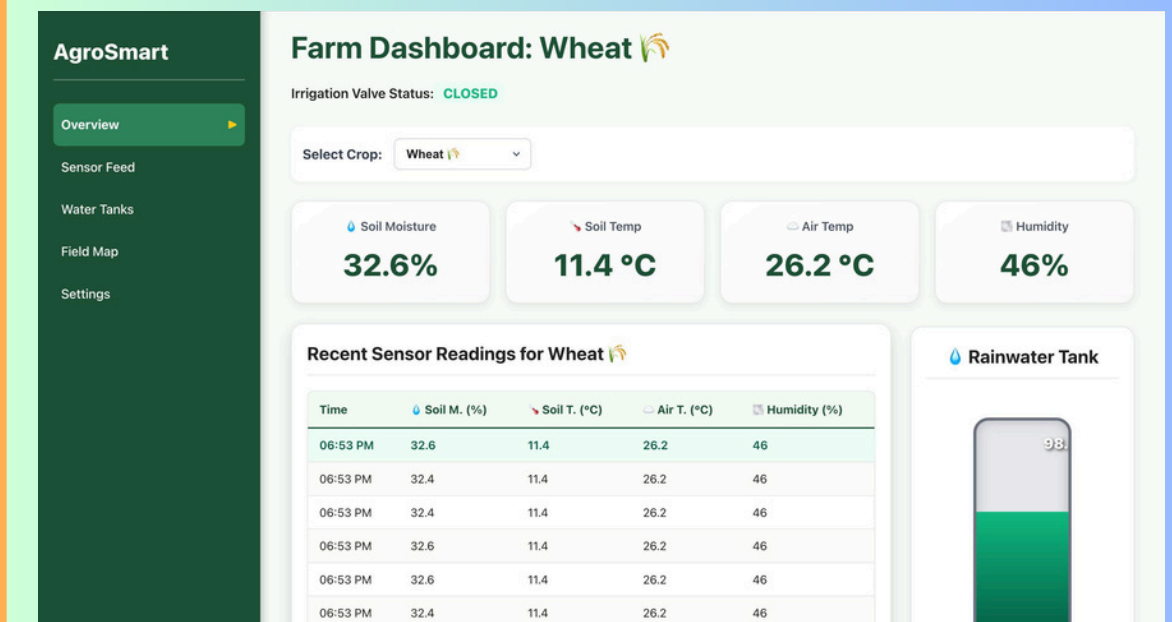
Flowchart



Technology used

- Sensors
- Microcontrollers
- Web dashboard
- SMS
- Solar power

Web dashboard



Analysis of feasibility of idea

- Proven, **low-cost IoT** and **solar technologies** are suitable for rugged hilly terrain and can **automate** irrigation efficiently using locally constructed percolation ponds.
- Low setup costs with a quick payback (**1–2** crop seasons) through higher yields (increase upto **20 - 30%**) and reduced input expenses make the system financially viable for small farmers.
- The system aligns with Sikkim's organic, terrace-based farming culture and is easily adoptable by farmers through simple **SMS-based alerts**, encouraging widespread acceptance.

Potential Challenges and Risks

- We need **constant power** supply while watering the fields which cannot be given during the nights.
- **Hardware maintenance** is also difficult as even conditions in hilly region may damage the components giving irregular values.

Strategies to overcome the Challenges

- Install a **battery storage system** along with the solar panels to store excess power generated during the day. This ensures the system (especially **pumps** and **valves**) has sufficient energy to run at night or during cloudy days
- Use **rugged**, industrial-grade sensors designed for agriculture in harsh environments, and place them in protective, weatherproof enclosures (**IP65** or higher).

BENIFITS OF SOULUTION

- Reliable year-round water supply despite uneven rainfall.
- Simple mobile/SMS alerts → easy to use even with low digital literacy.
- Increased crop productivity → improved food security & income.

🌱 Social :-

- Increases a farmer's exposure to newer technologies.
- Promotes adoption of sustainable farming in remote villages.

💰 Economic :-

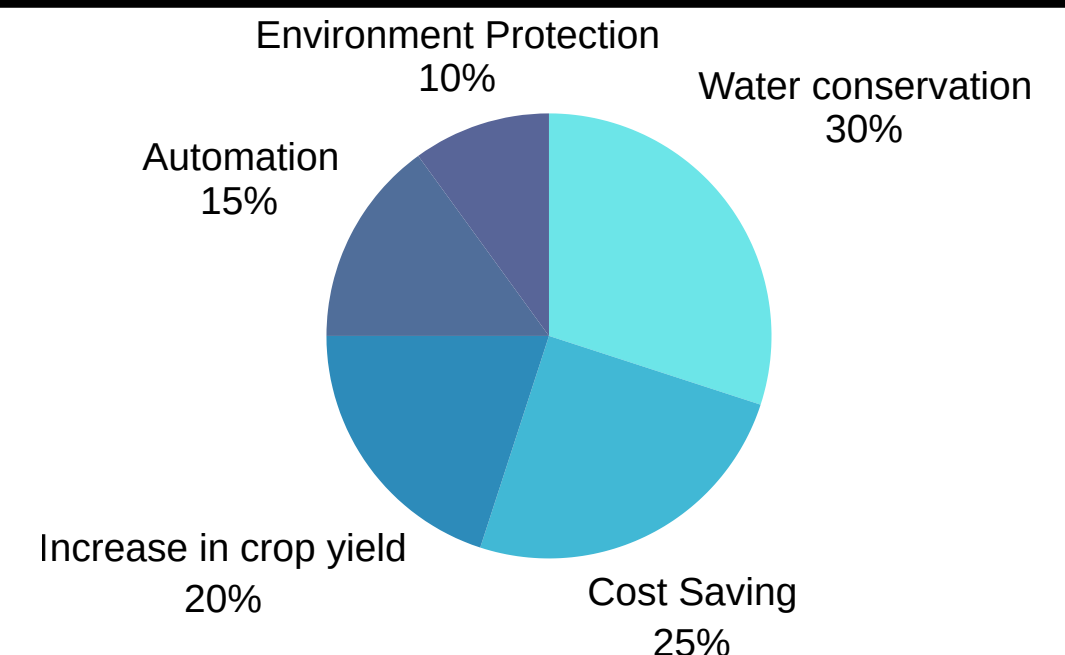
- 20–30% yield improvement with optimized water use.
- Lower maintenance costs(water,labour,electronic components).

🌍 Environmental :-

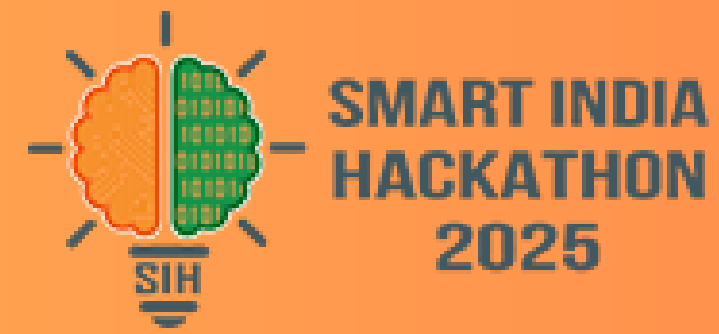
- Saves water by 30–40% through precise irrigation.
- Reduces soil erosion and nutrient runoff.
- Strengthens climate resilience in fragile hilly ecosystems.

Targeted Audience

- Small and Marginal farmers
- Farmer Producer Organizations (FPOs) and Cooperatives
- Government Agriculture Departments
- Rural Development Agencies



RESEARCH AND REFERENCES



- Analysing Challenges and Strategies in Land Productivity in Sikkim
- Hill Agriculture: Challenges and Opportunities
- Agriculture in Hilly and Mountainous Landscapes: Threats, Monitoring and Sustainable Management
- Studying ARDUINO